Portfolio Project Brief

Cybersecurity and Emerging Trends in Technology
Advanced Software Engineering - Module 3

How to use this brief:

Read this document in full and bring any questions you have on the assessment to either a workshop or the next 1:1 with your instructor. **Advance in your project journey by completing milestones**, which are progress checks on tasks that build to your overall output. Milestone submission and reviews will be part of your workshop and independent work. **Self-assess your progress** against the assessment criteria table while completing your project.

Module Title	Cybersecurity and Emerging Trends in Technology		
Format	Written Proposal	**Deadline**	On MyMultiverse
Length	3000 words	**Feedback Due**	Within 20 working days of submission date
Intended Audience	Non-technical Senior Leaders in your organization (but delivered to your instructor)	**Weighting**	100%
Project Title	Preparing and protecting an organization's information security		
Project Learning Outcomes	1. Research and explain how computer and/or data network(s) are designed, managed and secured. (K12, K16) 2. Design a simple digital and/or data network. (S12) 3. Evaluate a digital and/or data network for common vulnerabilities and risks. (K11, S9) 4. Propose a basic solution to increase the security of a digital and/or data network. (B4, B7)		
Project Milestones	**Milestone 1**: Evaluate and document potential security vulnerabilities in your organization's network and data management systems (weeks 1-3). **Milestone 2**: Develop a detailed security enhancement proposal for the network and data management systems (weeks 4-5). **Milestone 3:** Propose a pilot security enhancement, evaluate its effectiveness, and prepare a comprehensive impact report (by week 10)		

Feedback	Written feedback and a grade will be given within 20 working days after the deadline. Opportunity for further discussion will be available during individual coaching sessions.

Why is this project important to me and my organization?

This project is crucial for both your growth and the organization's security. It showcases your ability to manage and secure networks, conduct risk assessments, and propose resilience strategies, demonstrating your expertise in the latest cybersecurity trends. For the organization, it means a thorough evaluation of endpoint security, identifying vulnerabilities, and suggesting improvements to tackle emerging technology risks. Your work will strengthen the cybersecurity strategy and promote digital innovation and resilience.

What do I need to do for this project?

In this project, you will enhance your organization's network security by evaluating current protocols and identifying vulnerabilities using the NIST Cybersecurity Framework. Your tasks include creating an inventory of network endpoints, analyzing existing security measures, and proposing improvements. You will develop a detailed Network Endpoint Security Enhancement Proposal, which includes strategic recommendations and an impact assessment to demonstrate the benefits of your enhancements. This project showcases your ability to design secure network infrastructures, mitigate cyber threats, and communicate complex security strategies effectively to both technical and non-technical stakeholders.

Final Project Output

Your **Network Endpoint Security Enhancement Proposal** will encompass several detailed sections aimed at strengthening your network's defenses. Each section below is designed to guide you through the steps necessary for a thorough evaluation and enhancement of your network security. While suggested word counts are provided to maintain focus, they are flexible and should be adjusted as needed to comprehensively address all aspects of the security enhancement

Network Endpoint Security Enhancement Proposal Structure

Section	Content covered
1. Overview (500 words)	**Cybersecurity Significance:** Emphasize the strategic importance of cybersecurity in addressing the business implications of emerging technologies like cloud computing, IoT, and AI. Highlight how these technologies, while enhancing capabilities, also introduce new vulnerabilities. **Role and Project Impact:** Outline your responsibilities in managing network security and describe this project's alignment with your professional growth. Stress the importance of enhancing security to prevent cyber threats and maintain organizational integrity.
2. Evaluation of current state (1000 words)	**Endpoint Inventory:** • Create an inventory of accessible network endpoints categorized by role, operating system, and significance, using advanced scanning tools.

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	• Create a basic network diagram that can include components like routers, switches, servers, and workstations. **Cybersecurity Analysis** • Provide an overview of existing accessible and relevant protection mechanisms such as antivirus, anti-malware, and EDR systems, encryption and access control. • Analyze the efficacy of these tools, highlighting their advantages and disadvantages. • Analyze the effectiveness of protections and IDS/IPS systems such as firewalls, VPNs and endpoint protection software. • Identify areas needing improvement, focusing on deficiencies that could expose endpoints to threats.
3. Digital and/or data Network Security proposal (1200 words)	• Based on your analysis, propose strategic enhancements to address identified vulnerabilities. • Research, evaluate and propose new security technologies such as advanced threat detection systems and robust encryption methods to effectively bridge current security gaps. • Discuss the scalability, reliability, and security of the overall network architecture enhancements, justifying your proposal. • Explain how you would communicate proposal details to technical and non technical stakeholders to gain buy in.
4. Proposal and Impact story (300 words)	• **Quantifying Impact:** Develop an impact story that illustrates the ROI of the proposed security enhancements. Highlight potential cost savings, revenue improvements, and the enablement of other critical organizational functions. • Guidance Use: Reference the Business Impact Story Guide to structure this narrative effectively, ensuring the benefits are clear and compelling.

Examples of Projects

- **Advanced Threat Detection and Response:**
  - Implement cutting-edge threat detection systems that offer real-time monitoring and response capabilities. Projects might include the deployment of next-generation firewalls and sophisticated intrusion detection and prevention systems, all supported by automated threat intelligence gathering.
- **Encryption and Data Protection:**
  - Focus on enhancing encryption standards for both data at rest and data in transit. Projects in this category involve upgrading existing encryption protocols and employing advanced encryption technologies to ensure data integrity and confidentiality across all network endpoints.
- **Unified Threat Management and Network Segmentation:**
  - Develop a project that integrates unified threat management systems capable of managing diverse network environments. This includes establishing network segmentation to isolate and protect sensitive data areas, effectively reducing the attack surface.
- **Behavioral Analytics and Anomaly Detection:**

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    - Deploy behavioral analytics tools to monitor network and user activities continuously. Such projects utilize machine learning algorithms to detect unusual patterns that may indicate a security threat, enabling proactive incident prevention.
- **Comprehensive Endpoint Protection:**
    - Strengthen endpoint security through comprehensive protection strategies that include regular vulnerability assessments, the deployment of endpoint protection platforms, and the integration of antivirus and anti-malware solutions tailored to specific security needs..

Project Milestones

Project milestones are completion points for the overall output of your module project. Please note that the project is intended to be a piece of applied work that develops over the course of the module, rather than a summative assessment to be completed in a short, intense period of time at the end of module learning.

NOTE: For step-by-step guidance on milestone completion, follow the Milestone activities for the module on your MyMultiverse account.

Check Your Durable Skills

In addition to the assessment criteria, all assessments across your apprenticeship will also be marked on your durable skills. Use this checklist to ensure you are submitting a high quality piece of work.

Clear communication.

I've clearly and concisely communicated my findings and demonstrated professional language, spelling and grammar, and adhered to the word count. Please note that any referencing should be included at the end of the slide show, please refer to the style guide on Applied for further guidance.

- **Thinking strategies.** I've engaged in the critical analysis of available facts, evidence, observations, and arguments to form a judgment on what I think is important. Ultimately I've framed my analysis in a way that arrives at a solution.
- **Self-reflection.** I've considered feedback from previous work to inform the current submission. With this, I am engaging with feedback with a growth mindset in order to cultivate an attitude of continuous improvement.

How should I submit this Portfolio Project?

Your submission should be prepared as a single PDF file. Please make sure your work adheres to the MV referencing and style guide (see MyMultiverse). All work is checked for plagiarism using Turnitin.

Ensure that you meet the MV guidelines for academic integrity by considering the following:

- **Referencing:** When discussing someone else's work, clearly acknowledge the work by using in-text citations and a reference list adhering to the Harvard referencing style.
- **Word Count:** Any text exceeding the maximum word limit will not be marked. Text that adds to the word count includes all text except the reference list or appendices. All text in diagrams and in-text citations count towards the overall word limit.
- **Submission time:** All submissions must be made by mid-day on the specified deadline.

Please name your assignment submission file using the following format **<ULN_ModuleNumber_AssessmentNumber>** (e.g. 0012345_Module3_Assessment1).

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How will my assessment be marked?

Carefully review the assessment criteria as you complete milestones and outputs of your project.

Description	F	E	D	C	B	A
Research and Explain Network Design, Management, and Security Evidenced in: *2. Evaluation of the current state*	**Fails to conduct or articulate any research** related to the design, management, and security of computer and/or data networks.	**Conducts limited research** and **demonstrates a basic understanding** of network design, management, and security, **but cannot effectively explain key concepts** or best practices.	**Conducts research** and **understands key concepts** in network design, management, and security, but the explanation lacks clarity and depth in integrating these concepts. **Creates a basic network diagram**, but with omissions or inaccuracies.	**Conducts comprehensive research** and articulates a **clear understanding** of network design, management, and security, effectively explaining key concepts and best practices. Creates an accurate basic network diagram that includes components like routers, switches, servers, and workstations.	**Conducts comprehensive research,** provides a clear and **detailed explanation** of network design, management, and security, and enriches the discussion with insightful examples and best practices. **Creates a detailed and well-organized network diagram** that accurately represents the network endpoints and their roles.	**Conducts in-depth research**, offers original insights, and **exemplifies best practices** in network design, management, and security, setting a high standard for knowledge application and innovation. **Creates an exemplary network diagram** that is both comprehensive and precise, demonstrating advanced understanding of network architecture and endpoint management.

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Design a simple digital and/or data network. Evidenced in: *2. Evaluation of the current state* *3.Digital and/or data Network Security proposal*	**Unable to develop** a simple digital and/or data network design or explain design principles. ~~a~~	**Develops a simple** digital and/or data network design based on industry standards and regulations, **but struggles to explain** the design principles and considerations for building a secure digital and/or data network.	**Develops a novice** digital and/or data network design based on industry standards and regulations **and sufficiently explains** the design principles and considerations for building a secure digital and/or data network.	Develops a **well thought-out** digital and/or data network design based on industry standards and regulations and explains the design principles and considerations for building a secure digital and/or data network in a **clear and concise manner.**	**Develops a well thought-out** digital and/or data network design based on industry standards and regulations and **concisely explains** the design principles and considerations for building a secure digital and/or data network-with **additional insights and examples.**	Develops a **sophisticated** digital and/or data network design based on industry standards and regulations and **provides original insights and examples**.
Evaluate a digital and/or data network for common vulnerabilities and risks. Evidenced in: *2. Evaluation of the current state*	**Unable to conduct** a vulnerability assessment of a digital and/or data network or analyze the findings and identify the common vulnerabilities and risks.	**Partially conducts** a vulnerability assessment of a digital and/or data network using industry-standard tools, but **struggles to analyze** the findings and identify the common vulnerabilities and risks.	**Conducts a sufficient** vulnerability assessment of a digital and/or data network using industry-standard tools, **analyzes the findings, and identifies** the common vulnerabilities and risks.	**Conducts a sufficient** vulnerability assessment of a digital and/or data network using industry-standard tools, **analyzes the findings, and identifies** the common vulnerabilities and risks in a **clear and concise manner.**	**Conducts a thorough** vulnerability assessment of a digital and/or data network using industry-standard tools, a**nalyzes the findings, and identifies** the common vulnerabilities and risks in a **clear and concise manner, with additional insights and examples.**	**Conducts an in-depth** vulnerability assessment of a digital and/or data network using industry-standard tools **analyzes the findings, and provides original insights and examples.**

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Propose a basic solution to increase the security of a digital and/or data network. Evidenced in: *3.Digital and/or data Network Security proposal* *4.Proposal and Impact story*	**Unable to develop a basic solution** to address the identified vulnerabilities and risks or explain how the proposed solution aligns with industry standards and regulations.	**Develops a basic solution** to address the identified vulnerabilities and risks, but **struggles to explain how** the proposed solution aligns with industry standards and regulations.	**Develops a sufficient solution** to address the identified vulnerabilities and risks and **vaguely explains how** the proposed solution aligns with industry standards and regulations.	**Develops a well thought-out solution** to address the identified vulnerabilities and risks and **thoroughly explains how** the proposed solution aligns with industry standards and regulations in a **clear and concise manner.**	**Develops a sophisticated solution** to address the identified vulnerabilities and risks and **thoroughly explains how** the proposed solution aligns with industry standards and regulations in a **clear and concise manner, with additional insights and examples.**	**Develops an innovative solution** to address the identified vulnerabilities and risks and provides original insights and examples.

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